

UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Artificial Intelligence

TOPIC 7:
LINEAR REGRESSION

SAIA 2113 – Machine Learning
Session 2 2024/2025

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Innovating Solutions

Topic 7: Linear Regression

- What is linear regression?
- Loss
- Gradient descent
- Hyperparameters

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What is linear regression?

- Linear regression is a statistical technique used to find the relationship between variables. In the context of machine learning, linear regression finds the relationship between features and a label.

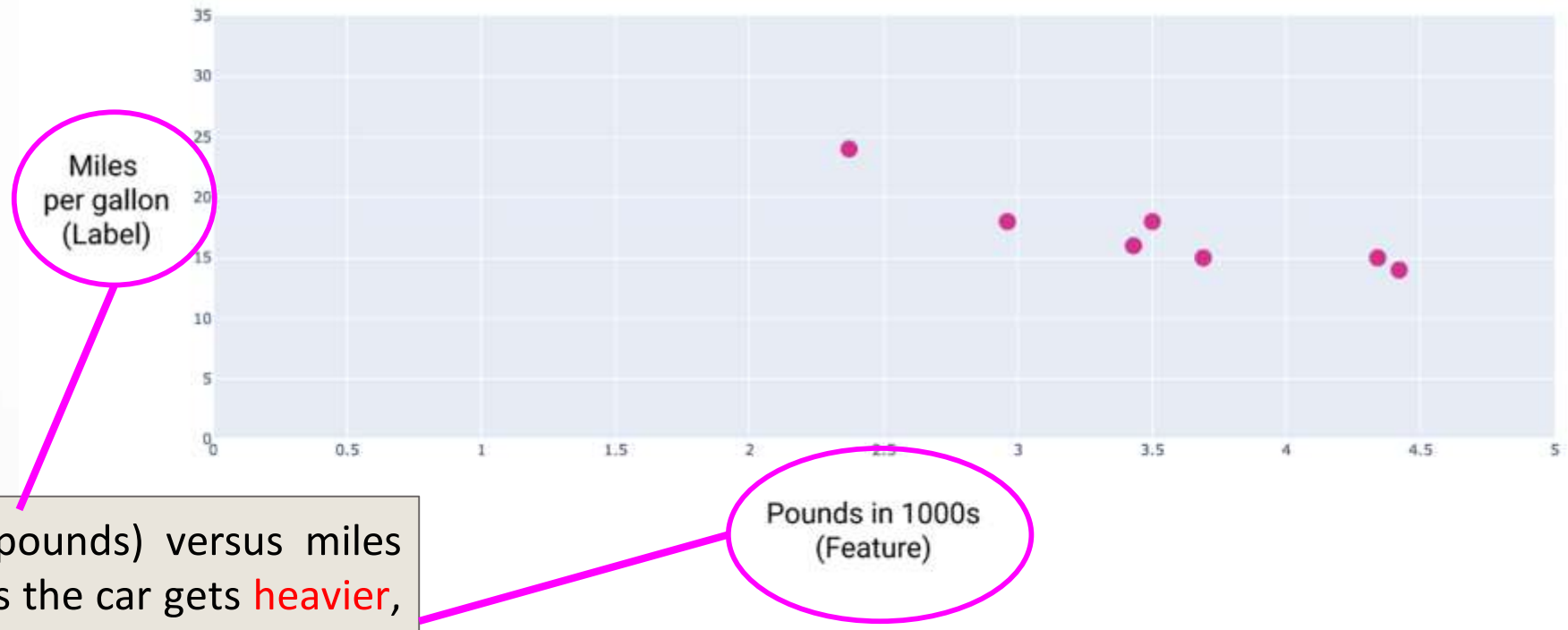
- For instance, assume that we want to predict a car's fuel efficiency in miles per gallon (mpg) based on how heavy the car is and the following dataset is used.

Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
3.44	18
3.43	16
4.34	15
4.42	14
2.37	24

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

If we plotted points from the table, we'd get the following graph:

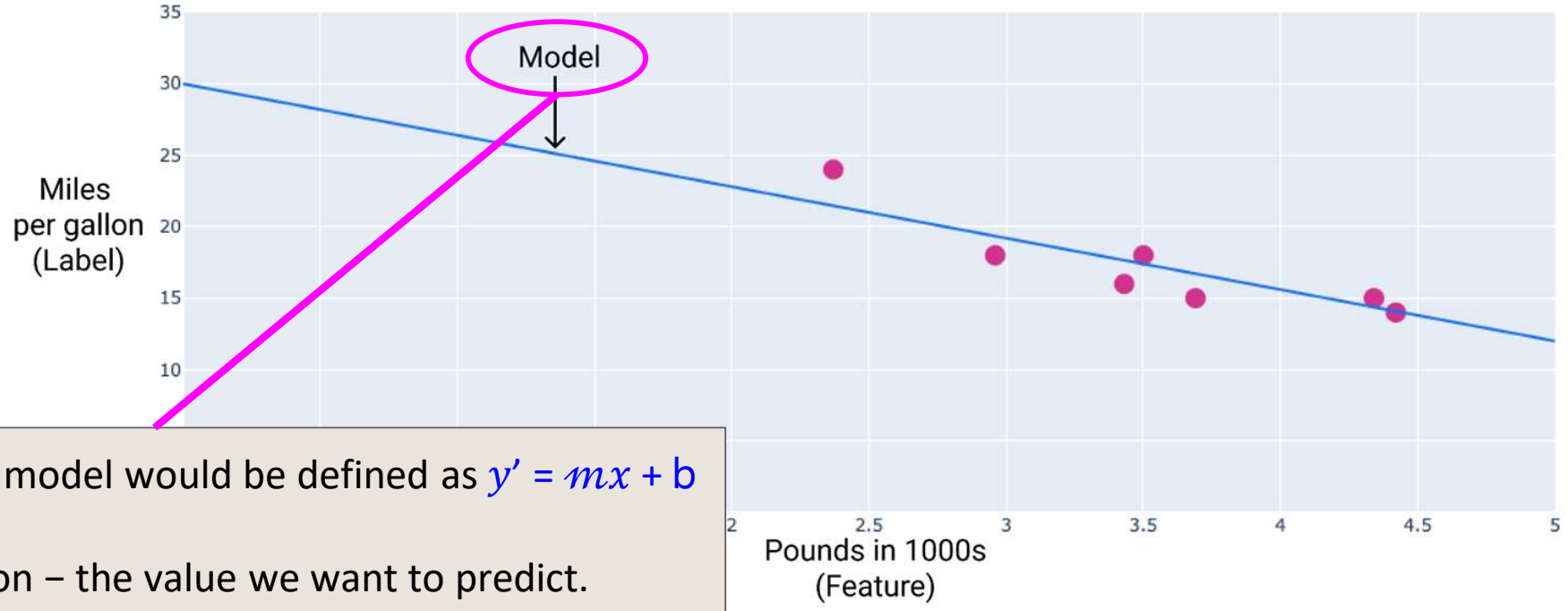


Car heaviness (in pounds) versus miles per gallon rating. As the car gets heavier, its miles per gallon rating generally decreases.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

We could create our own model by drawing a best fit line through the points:



In algebraic terms, the model would be defined as $y' = mx + b$

- y is miles per gallon – the value we want to predict.
- m is the slope of the line.
- x is pounds – our input value.
- b is the intercept.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

In machine learning, the equation for a linear regression can be written as follows:

$$y' = b + w_1 x_1$$

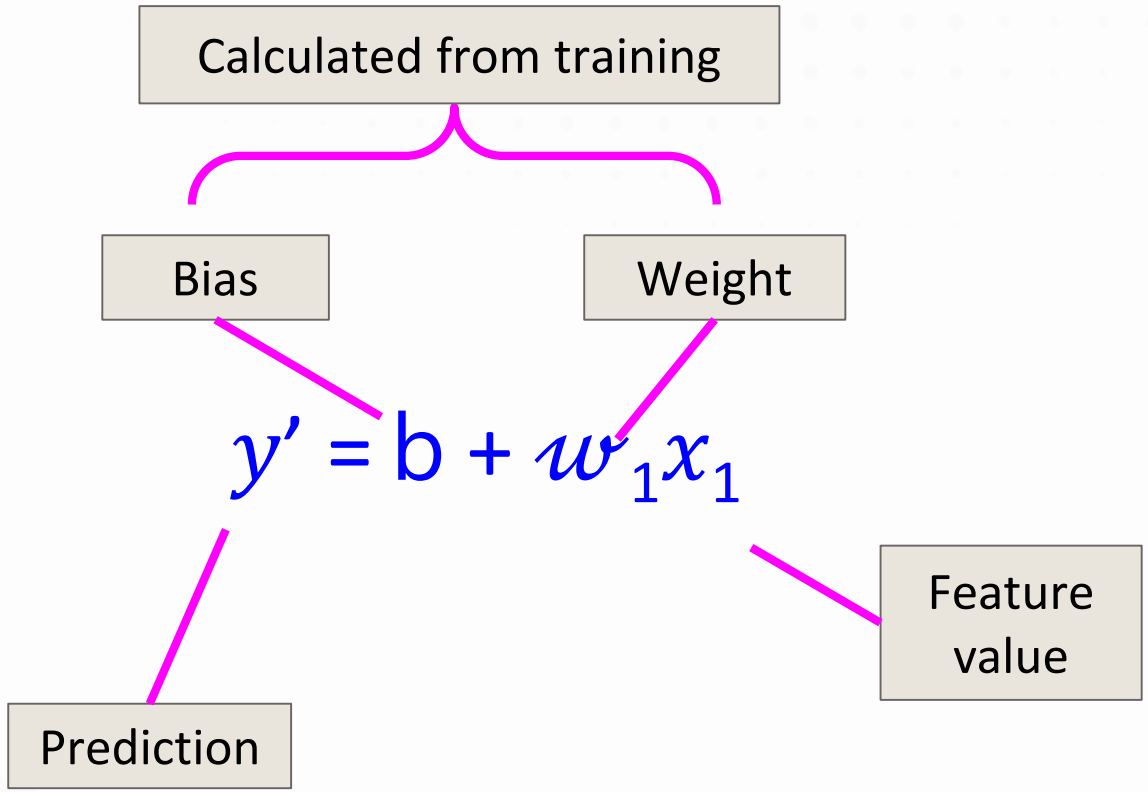
where:

- y' is the predicted label − the output.
- b is the bias of the model. Bias is the same concept as the y-intercept in the algebraic equation for a line. In ML, bias is sometimes referred to as w_0.
- w_1 is the weight of the feature. Weight is the same concept as the slope m in the algebraic equation for a line. Weight is a parameter of the model and is calculated in training.
- x_1 is a feature − the input.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

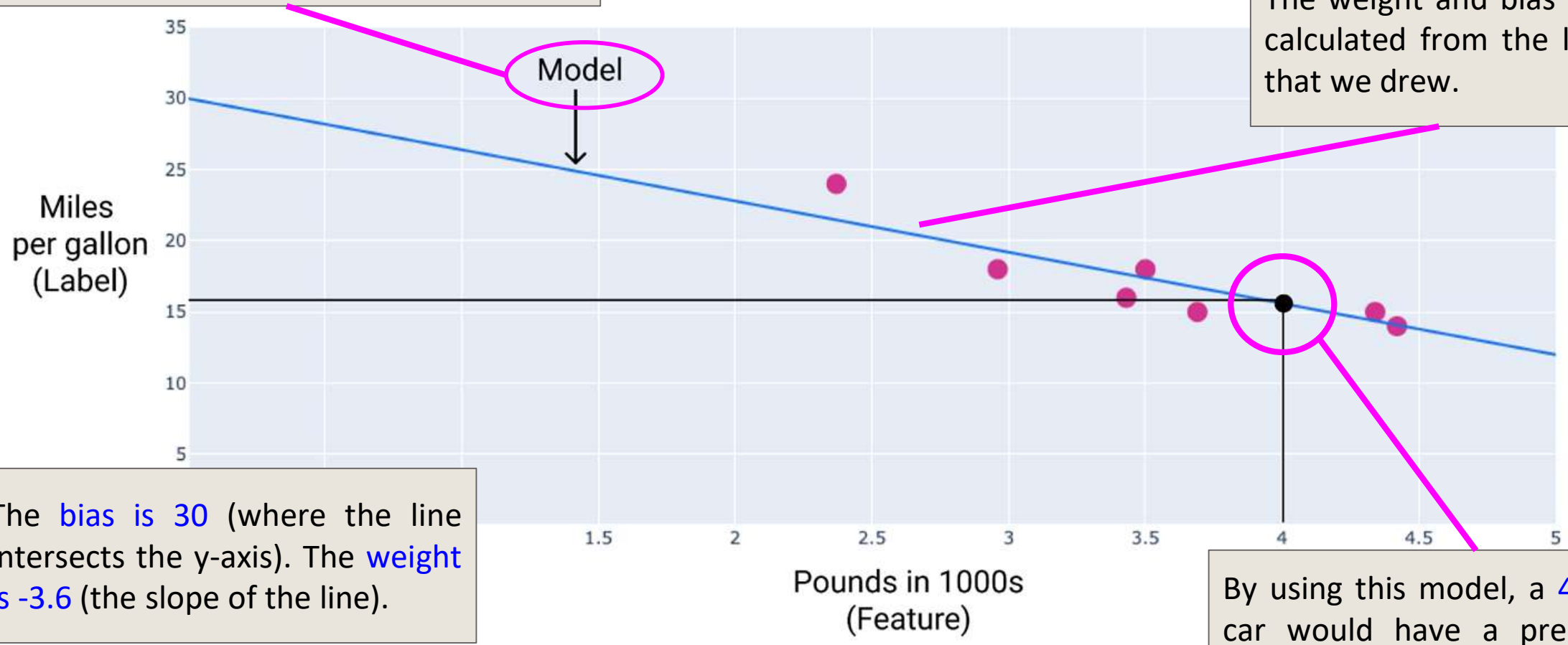
During training, the model calculates the weight and bias that produce the best model.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

The model used to make predictions would be defined as $y' = 30 + (-3.6)(x_1)$



The weight and bias are calculated from the line that we drew.

The bias is 30 (where the line intersects the y-axis). The weight is -3.6 (the slope of the line).

By using this model, a 4000-pound car would have a predicted fuel efficiency of 15.6 miles per gallon.

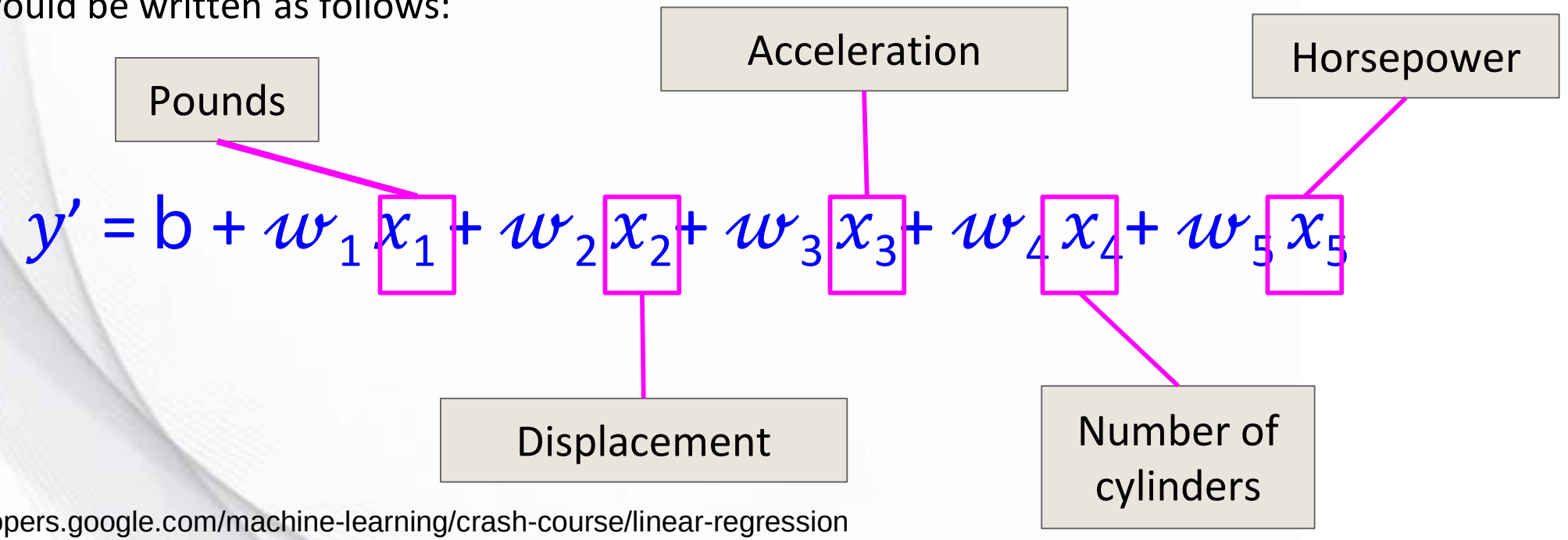
Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?

Models with multiple features

A more sophisticated model might rely on multiple features. Each of the features have separate weight (w_1, w_2, etc). For example, a model could rely on the following five features: (1) pounds; (2) engine displacement; (3) acceleration; (4) number of cylinders; and (5) horsepower.

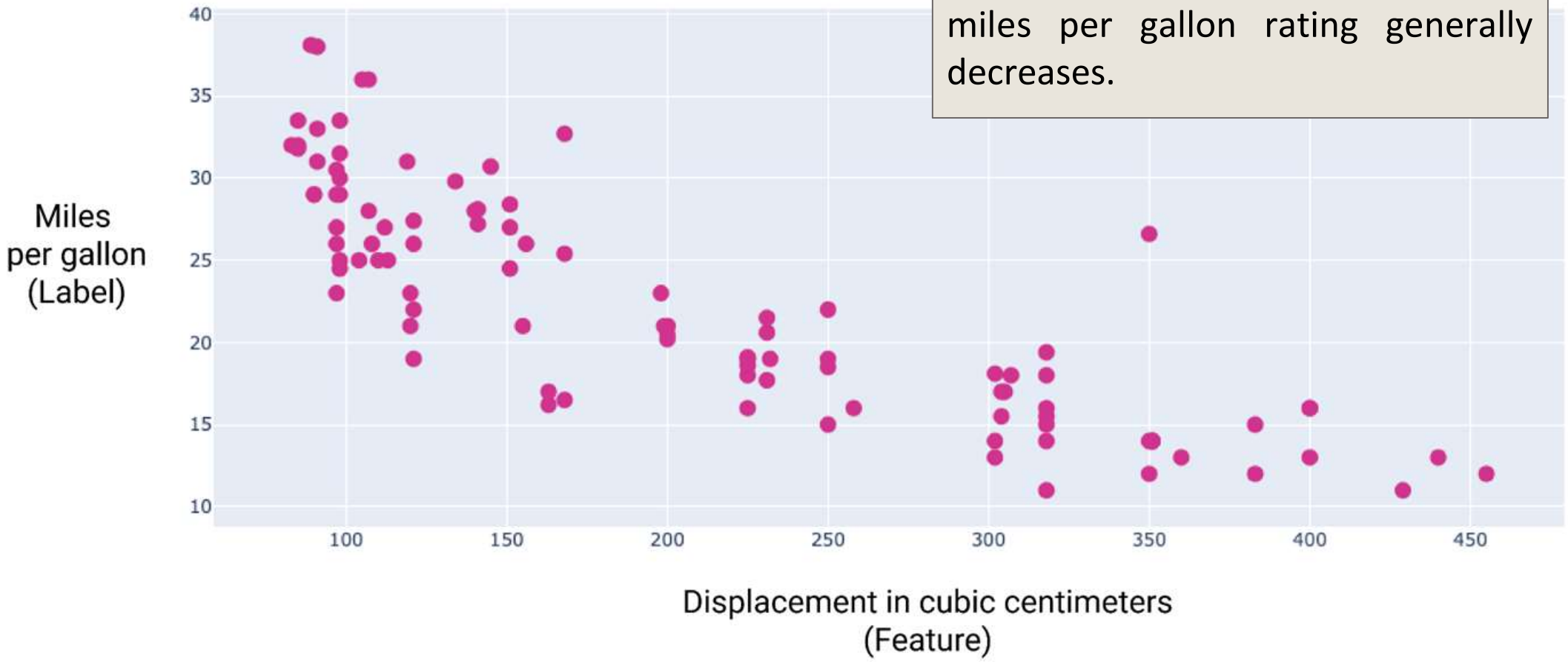
The model would be written as follows:



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

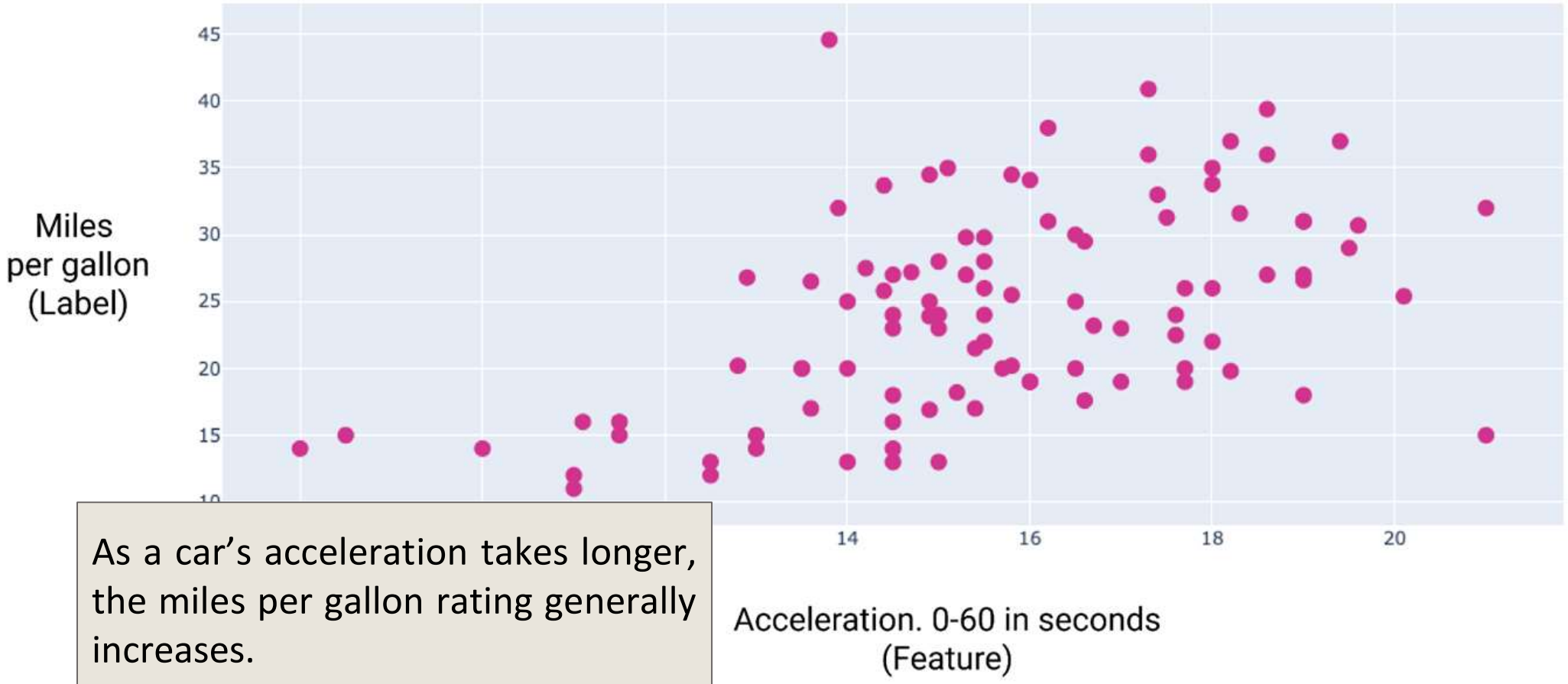
What is linear regression?

As a car's engine gets bigger, its miles per gallon rating generally decreases.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

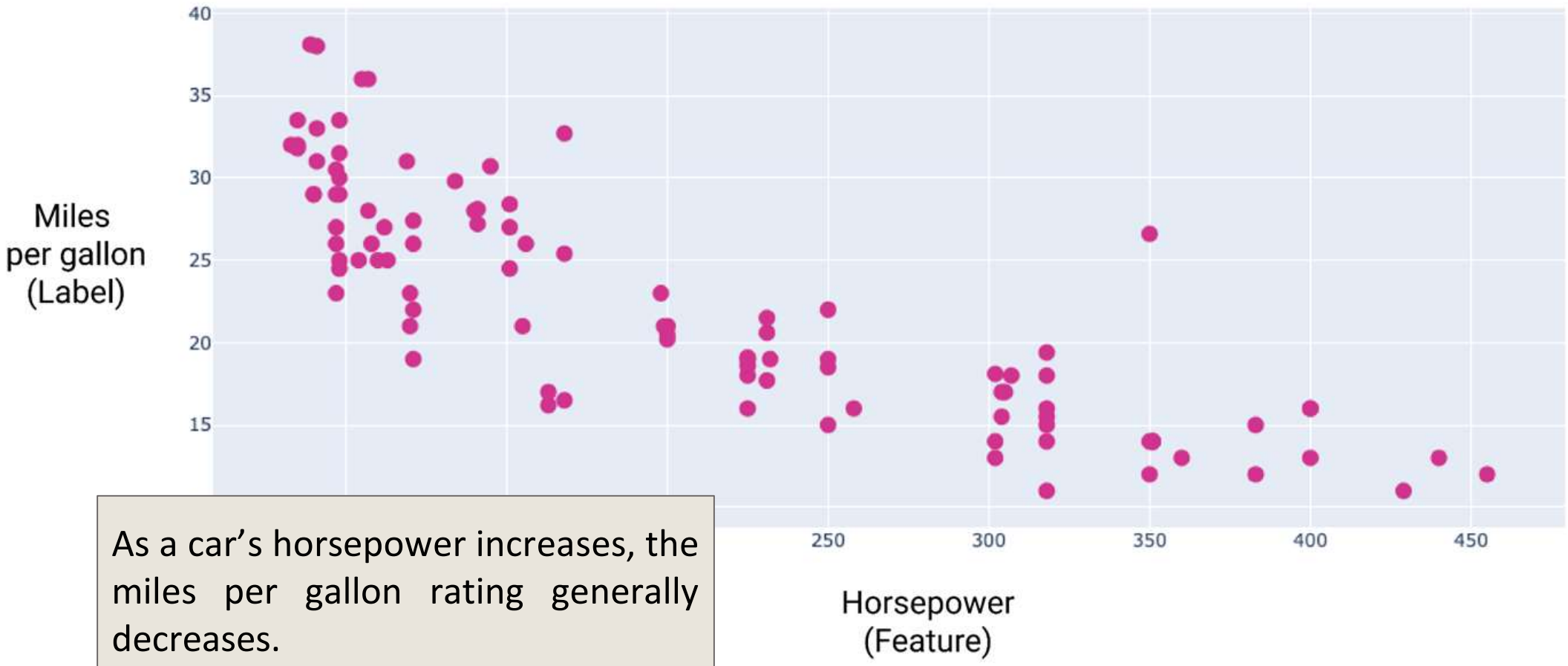
What is linear regression?



As a car's acceleration takes longer, the miles per gallon rating generally increases.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

What is linear regression?



As a car's horsepower increases, the miles per gallon rating generally decreases.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Topic 7: Linear Regression

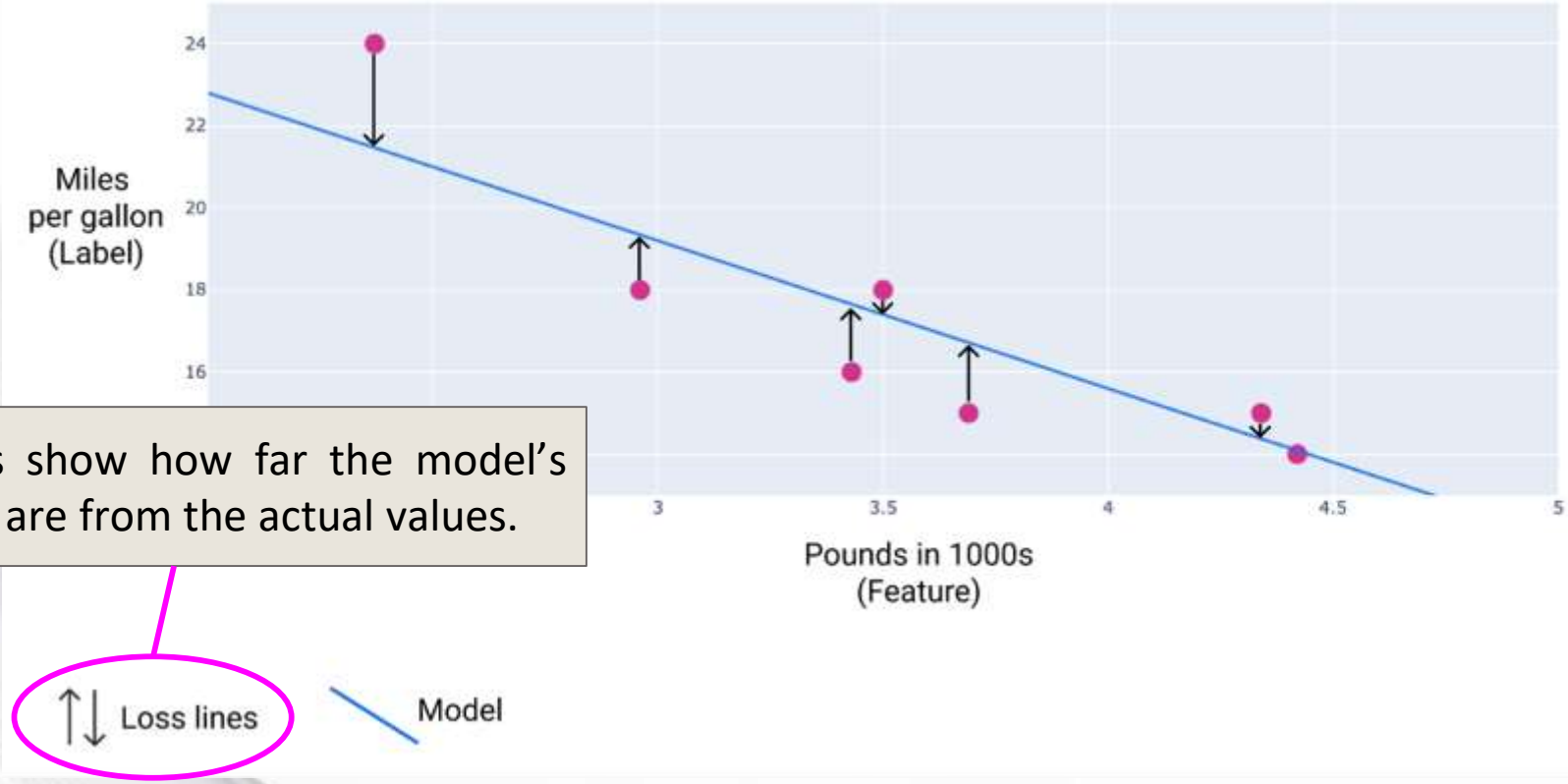
- What is linear regression?
- Loss
- Gradient descent
- Hyperparameters

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Loss

Loss is a numerical metric that describes how wrong a model's prediction are.

Loss measures the distance between the model's predictions and the actual labels. The goal of training a model is to minimize the loss, reducing it to its lowest possible value.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Distance of loss

For example, if a model predicts 2, but the actual value is 5, we can ignore that the loss is negative −3 (2 − 5 = −3). Instead, we consider that the distance between the values is 3. Thus, all methods for calculating loss remove the sign.

The most two common methods to remove the sign are the following:

- Take the absolute value of the difference between the actual value and the prediction.
- Square the difference between the actual value and the prediction.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Type of loss

There are four main types of loss:

Loss type	Definition	Equation
L_1 loss	The sum of the absolute values of the difference between the predicted values and the actual values.	$\sum \lvert \text{actual value} - \text{predicted value} \rvert$
Mean absolute error (MAE)	The average of L_1 losses across a set of examples.	$1 / N \sum \lvert \text{actual value} - \text{predicted value} \rvert$
L_2 loss	The sum of the squared difference between the predicted values and the actual values.	$\sum \lvert \text{actual value} - \text{predicted value} \rvert^2$
Mean squared error (MSE)	The average of L_2 losses across a set of examples.	$1 / N \sum \lvert \text{actual value} - \text{predicted value} \rvert^2$

When the difference between the prediction and label is large, squaring makes the loss even larger. When the difference is small (less than 1), squaring makes the loss even smaller.

It is recommended when processing multiple examples at once to average the losses by using MAE or MSE.

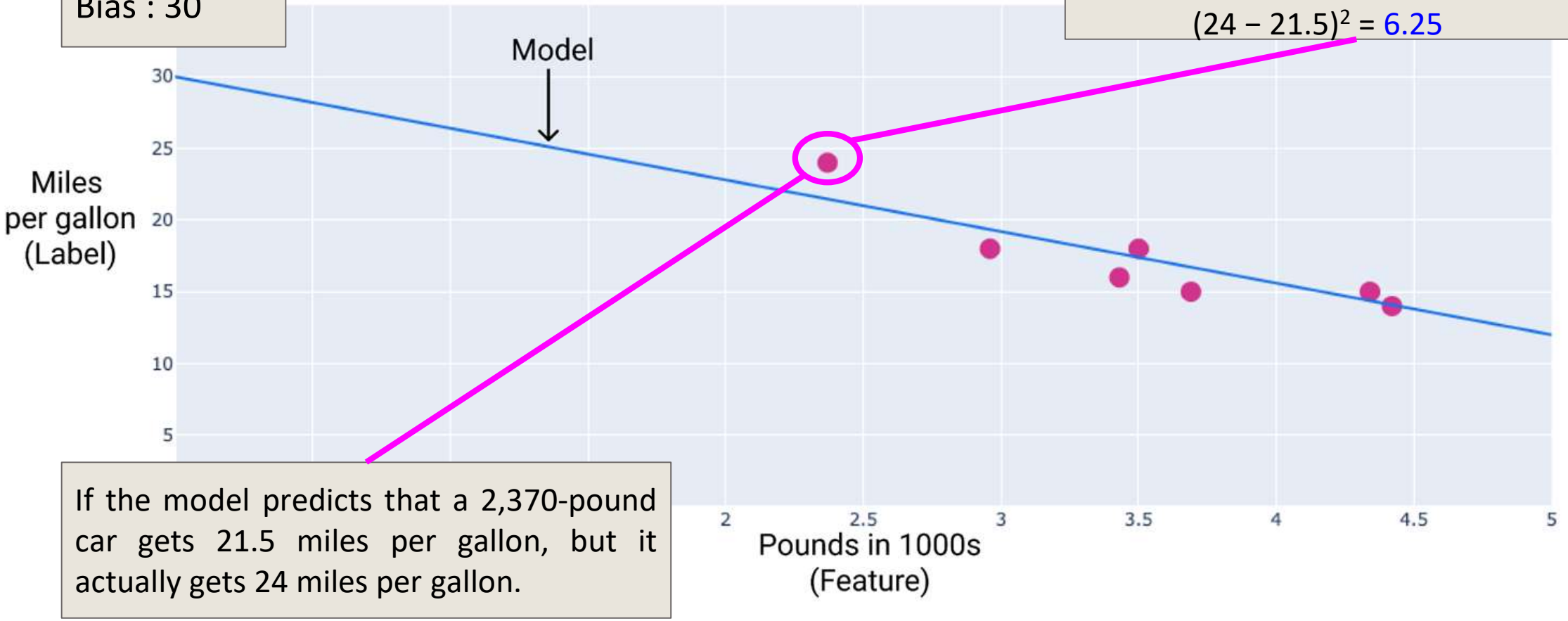
Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Calculating loss

Weight: -3.6
Bias : 30

L_2 loss = (actual value − predicted value)^2

$(24 - 21.5)^2 = 6.25$

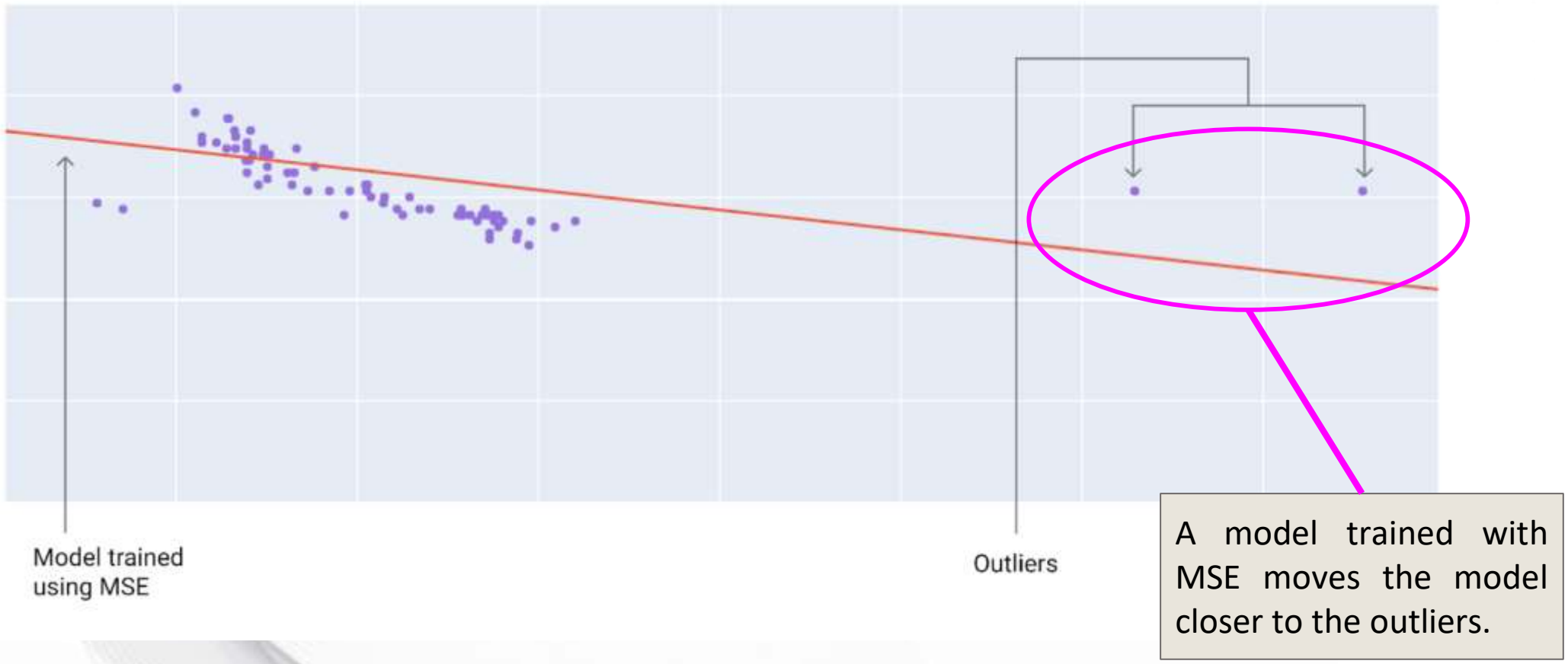


If the model predicts that a 2,370-pound car gets 21.5 miles per gallon, but it actually gets 24 miles per gallon.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

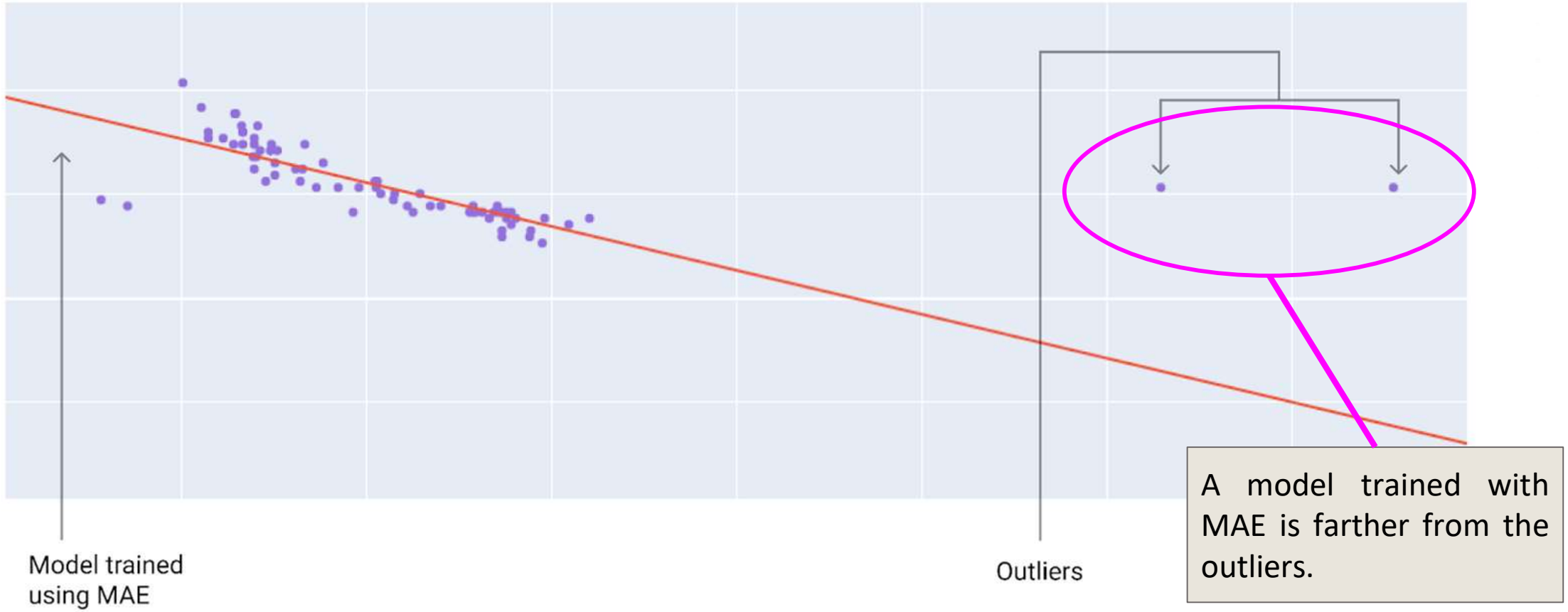
Choosing a loss

When choosing the best loss function, consider how you want the model to treat outliers.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Choosing a loss



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Choosing a loss

Note the relationship between the model and the data:

- **MSE.** The model is closer to the outliers but further away from most of the other data points.
- **MAE.** The model is further away from the outliers but closer to most of the other data points.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Topic 7: Linear Regression

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Gradient descent

Gradient descent is a mathematical technique that iteratively finds the weight and bias that produce the model with the lowest loss.

The model begins training with randomized weights and biases near zero, and then repeats the following steps:

1. Calculate the loss with the current weight and bias.
2. Determine the direction to move the weights and bias that reduce loss.
3. Move the weight and bias values a small amount in the direction that reduces loss.
4. Return to step one and repeat the process until the model can't reduce the loss any further.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Gradient descent

At a concrete level, the gradient descent steps can be followed by using a small dataset with seven examples for a car's heaviness in pounds and its miles per gallon rating:

Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
3.44	18
3.43	16
4.34	15
4.42	14
2.37	24

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Gradient descent

1. The model starts training by setting the weight and bias to zero:

Weight : 0
Bias : 0

$y = 0 + 0\ (x_1)$

1. Calculate MSE loss with the current model parameters:

$$\text{Loss} = \frac{(18-0)^2 + (15-0)^2 + (18-0)^2 + (16-0)^2 + (15-0)^2 + (14-0)^2 + (24-0)^2}{7}$$

Loss = 303.71

1. Calculate the slope of the tangent to the loss function at each weight and the bias:

Weight slope: −119.7

Bias slope: −34.3

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Gradient descent

4. Move a small amount in the direction of the negative slope to get the next weight and bias. For now, we'll arbitrarily define the "small amount" as 0.01:

New weight = old weight − (small amount * weight slope)
New bias = old bias − (small amount*bias slope)

New weight = 0 − (0.01) * (-119.7)
New bias = 0 − (0.01) * (−34.3)

New weight = 1.2
New bias = 0.34

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Gradient descent

Use the new weight and bias to calculate the loss and repeat. Completing the process for six iterations, we'd get the following weights, biases, and losses.

Iteration	Weight	Bias	Loss (MSE)
1	0	0	303.71
2	1.2	0.34	170.67
3	2.75	0.59	67.3
4	3.17	0.72	50.63
5	3.47	0.82	42.1
6	3.68	0.9	37.74

The loss gets lower with each updated weight and bias.

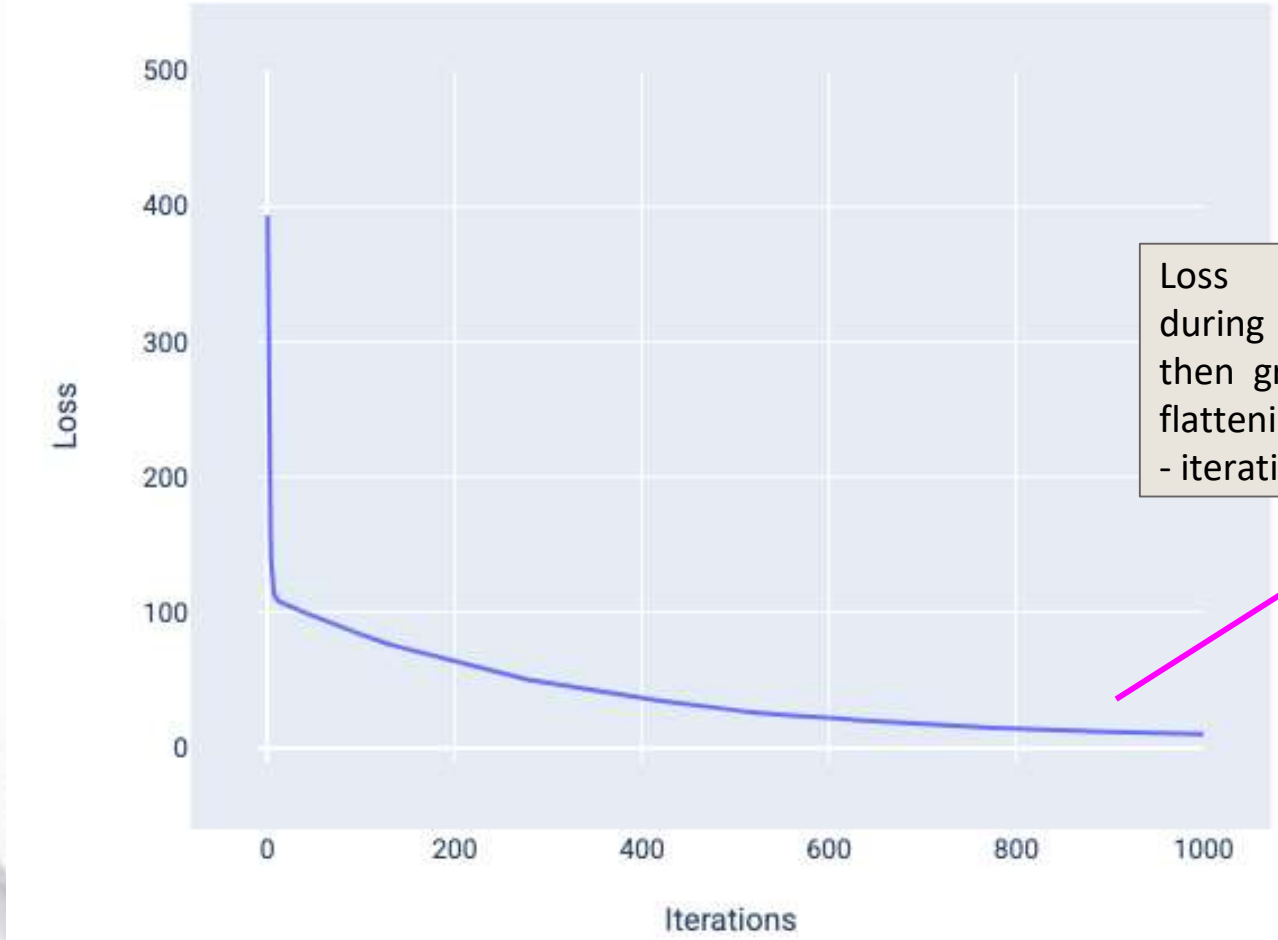
In this example, we stopped after six iterations. In practice, a model trains until it converges.

When a model converges, additional iterations don't reduce loss more because gradient descent has found the weights and bias that nearly minimize the loss.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Model convergence and loss curves

The loss curve are often observed to determine whether the model has converged during training.

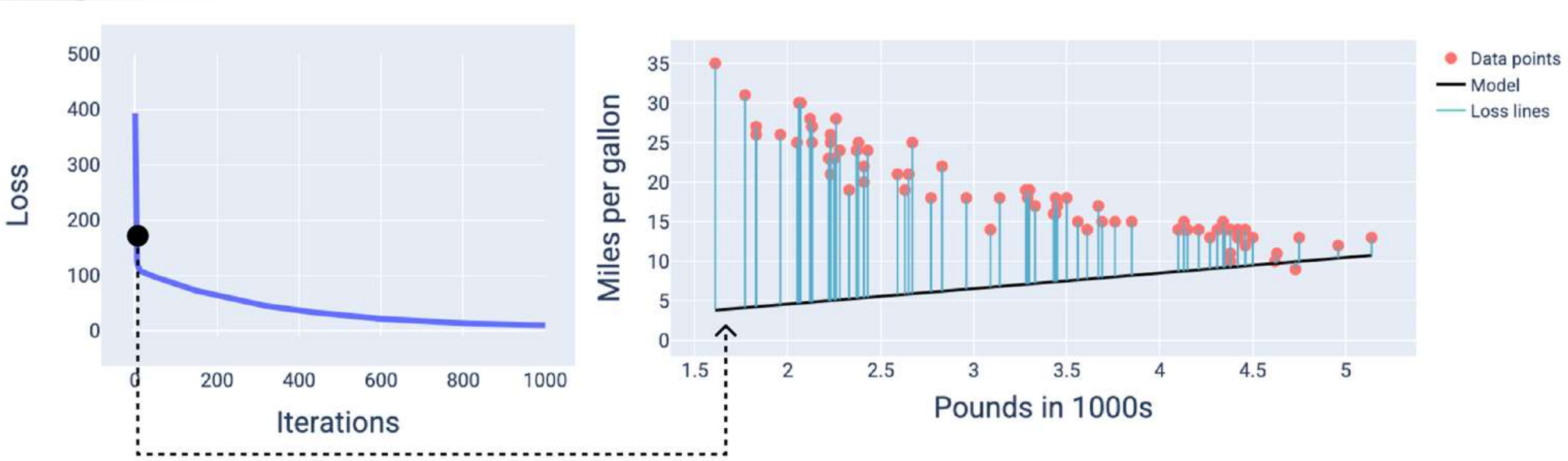


Loss dramatically decreases during the first few iterations, then gradually decreases before flattening out around the 1,000th - iteration mark.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Model convergence and loss curves

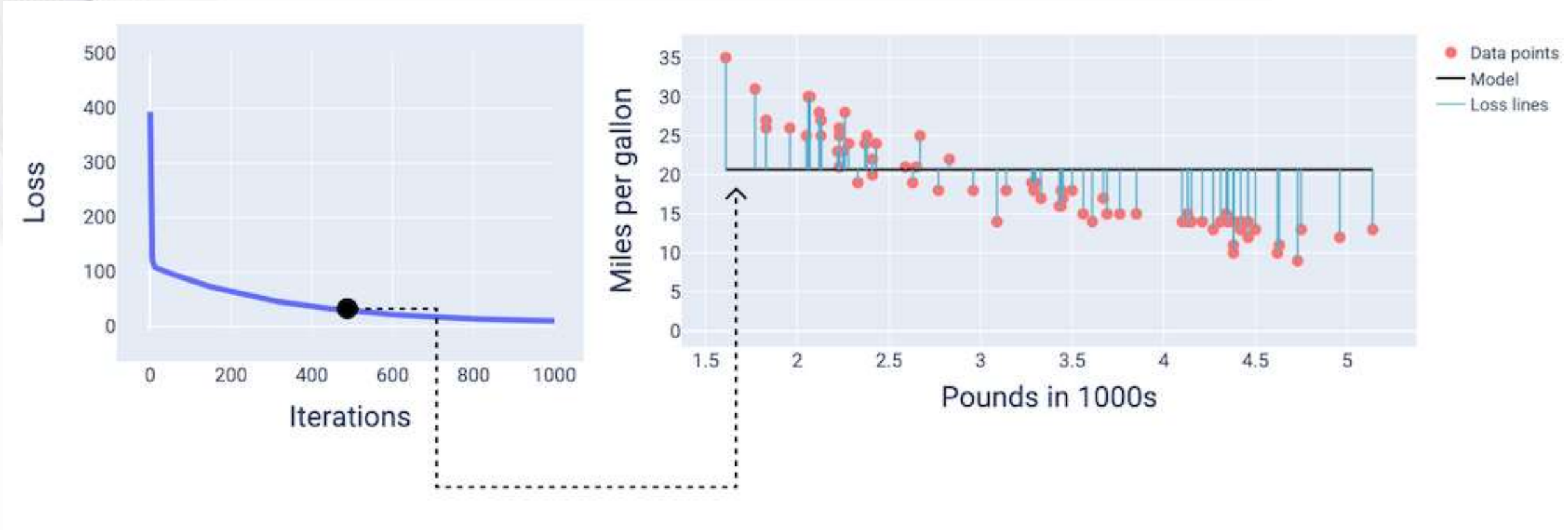
In the following figure, the second iteration indicates that the model would not be good at making predictions because of the high amount of loss.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Model convergence and loss curves

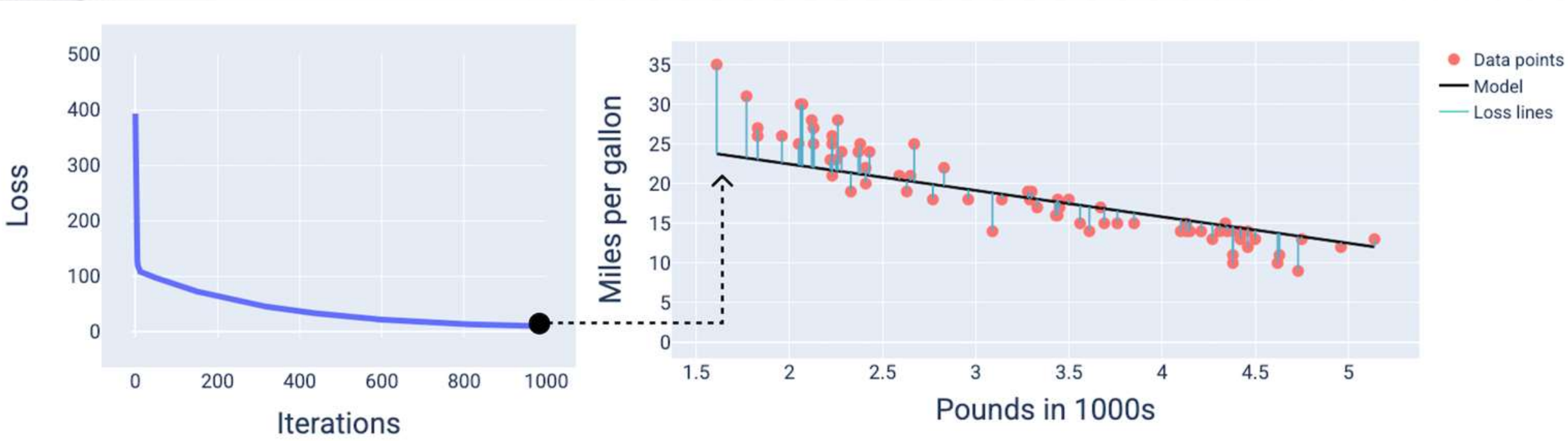
At around the 400th-iteration, we can see that gradient descent has found the weight and bias that produce a better model.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Model convergence and loss curves

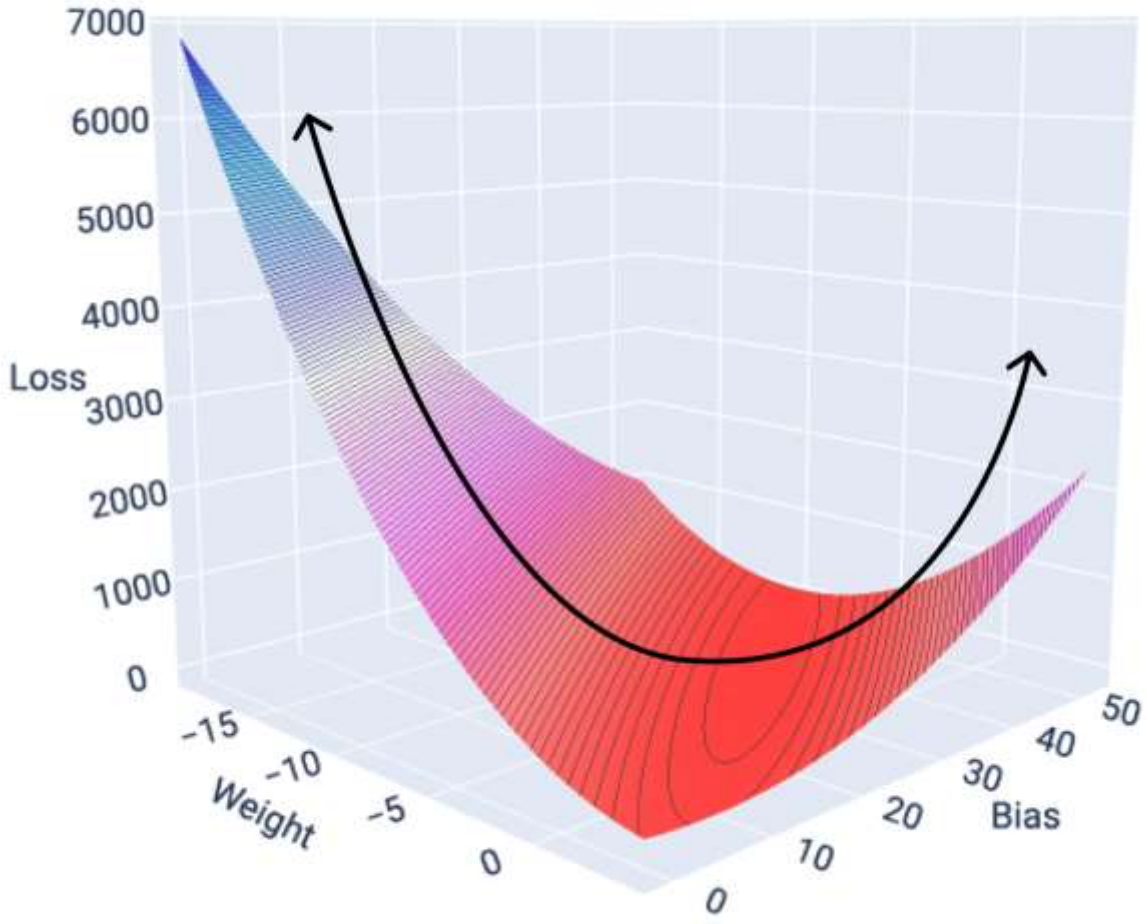
Around the 1000th-iteration, we can see that the model has converged, producing a model with the lowest possible loss.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

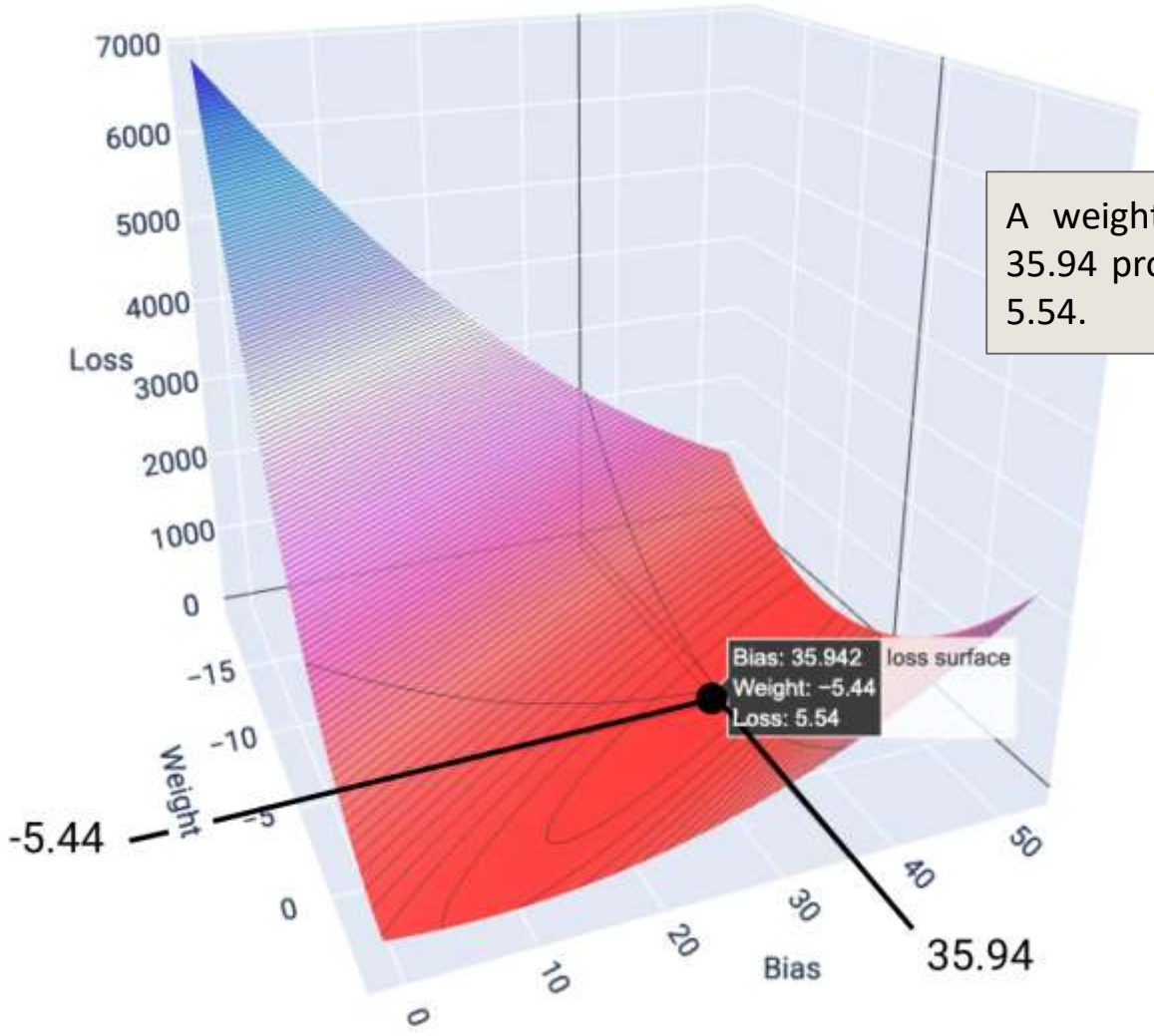
Convergence and convex functions

- The loss functions of linear models always produce a convex surface.
- If we graph the loss surface of a model with one feature, we can see its convex shape.
- Weight is on the x-axis, bias is on the y-axis, and loss is on the z-axis.



Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Convergence and convex functions

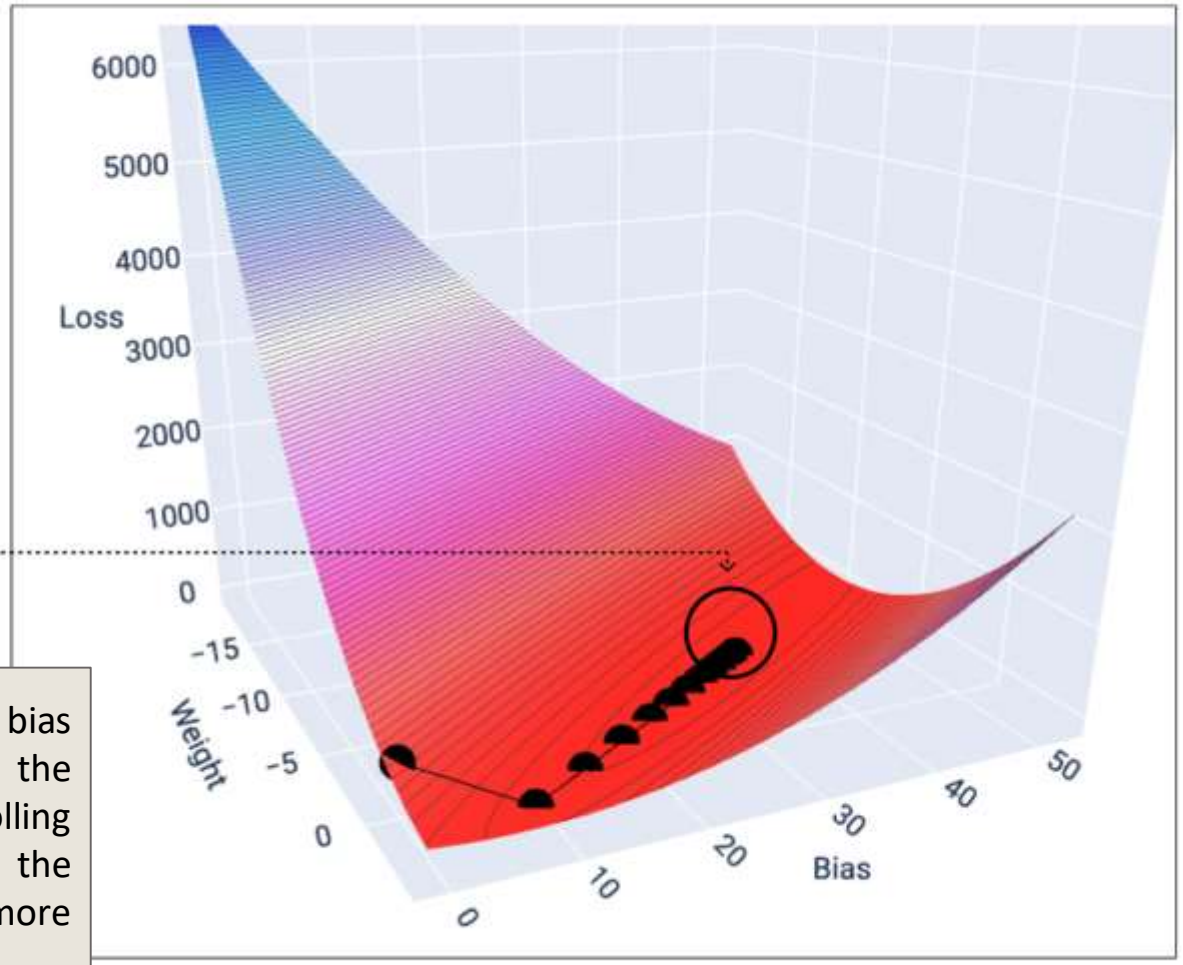


A weight of -5.44 and bias of 35.94 produce the lowest loss at 5.54.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Convergence and convex functions

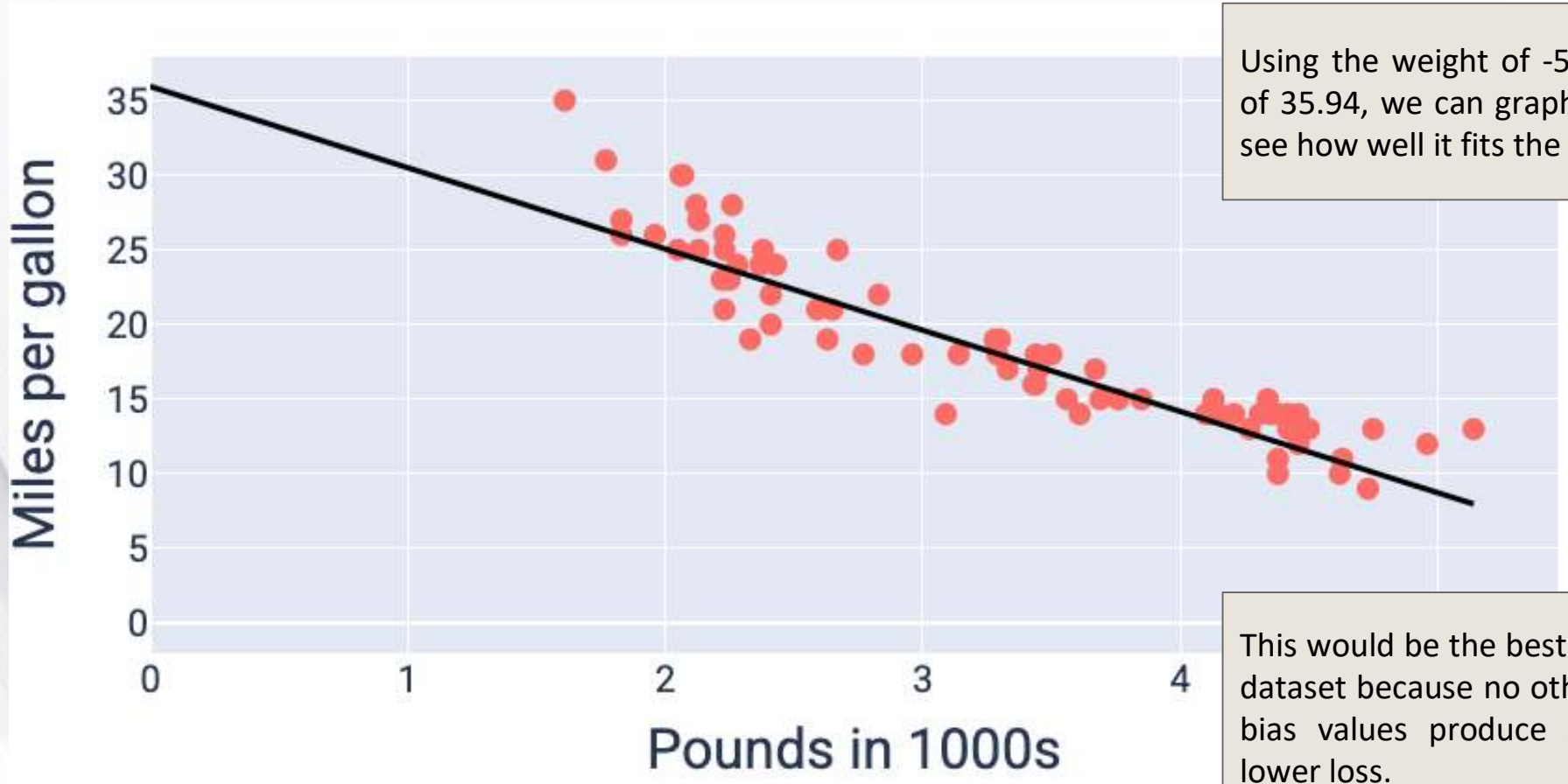
Linear regression models converge because the loss surface is convex and contains a point where the weight and bias have a slope that's almost zero.



If we graphed the weights and bias points during gradient descent, the points would look like a ball rolling down a hill, finally stopping at the point where there's no more downward slope.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Convergence and convex functions



Using the weight of -5.44 and a bias of 35.94, we can graph the model to see how well it fits the data.

This would be the best model for this dataset because no other weight and bias values produce a model with lower loss.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Topic 7: Linear Regression

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Hyperparameters

Hyperparameters are variables that control different aspects of training. There are three common hyperparameters:

- Learning rate
- Batch size
- Epochs

In contrast, parameters are the variables such as the weight and bias of the model. In other words, hyperparameters are values that you control. Parameters are values that the model calculates during training.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

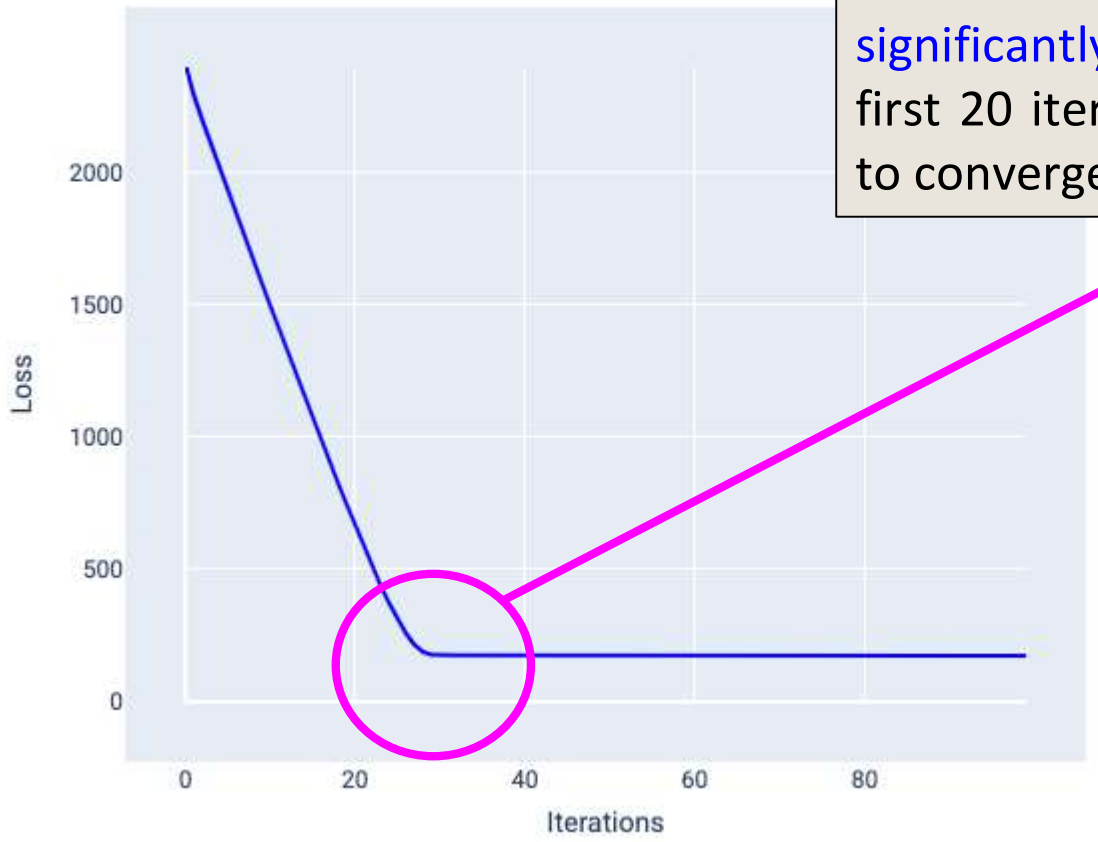
Learning rate

- Learning rate is a floating-point number you set that influences how quickly the model converges.
- If the learning rate is too low, the model can take a long time to converge.
- However, if the learning rate is too high, the model never converges, but instead bonus around the weights and bias that minimize the loss.
- The goal is to pick a learning rate that is not too high nor too low so that the model converges quickly.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Learning rate

The ideal learning rate helps the model to converge within a reasonable number of iterations.

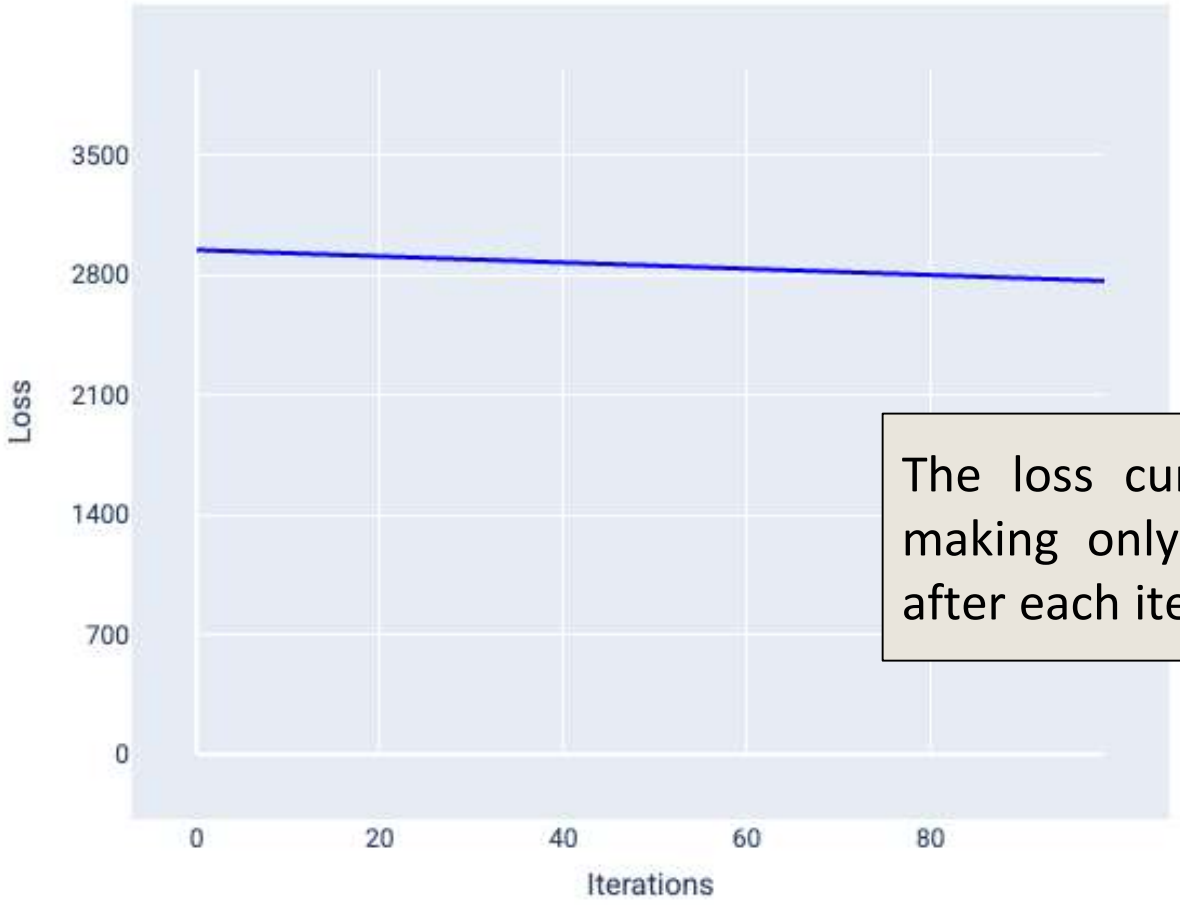


The loss curve shows the model significantly improving during the first 20 iterations before beginning to converge.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Learning rate

In contrast, a learning rate that is too small can take many iterations to converge.

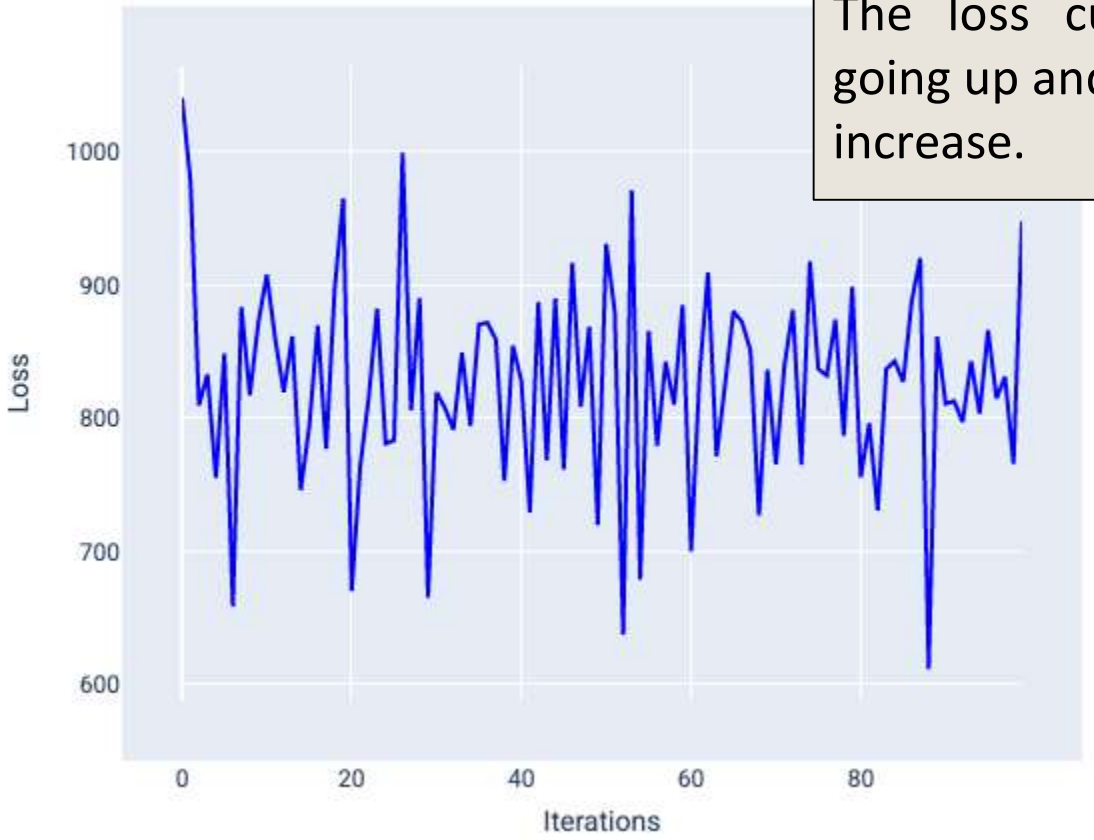


The loss curve shows the model making only minor improvements after each iteration.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Learning rate

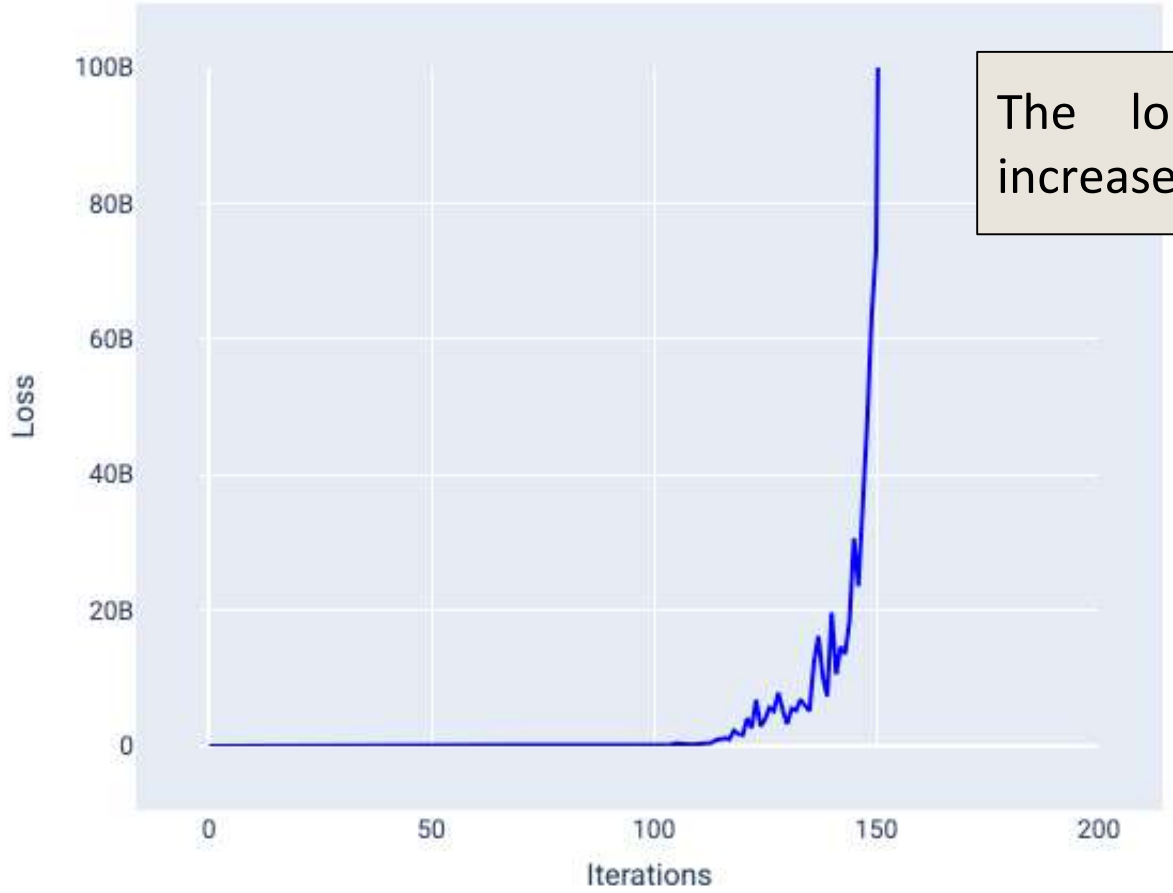
A learning rate that is too large never converges because each iteration either causes the loss to bounce around or continually increase.



The loss curve fluctuates wildly, going up and down as the iterations increase.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Learning rate



The loss curve drastically increases in later iterations.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

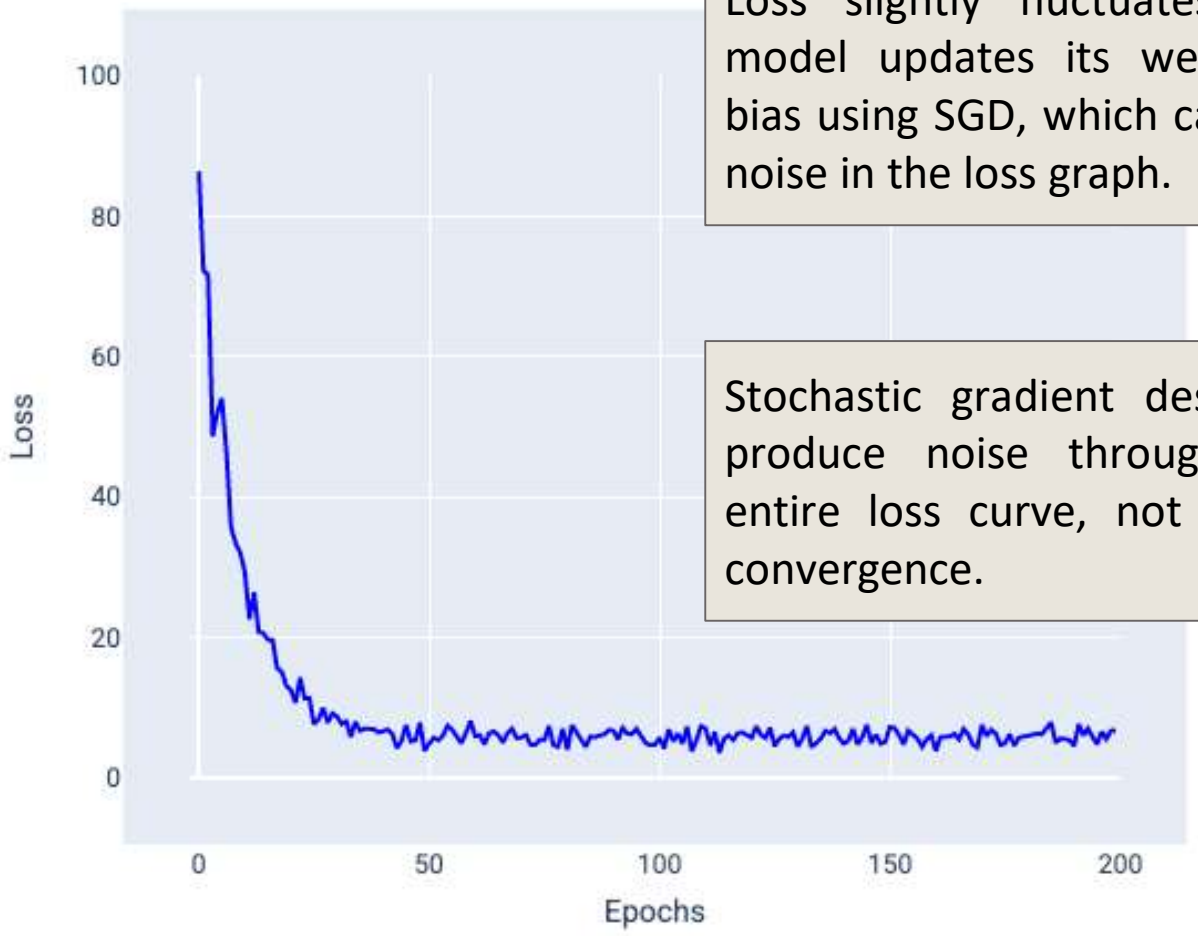
Batch size

- Batch size is a hyperparameter that refers to the number of examples the model processes before updating its weight and bias.
- When a dataset contains hundreds of thousands or even millions of examples, using the full batch isn't practical.
- Two common techniques to get the right gradient on average without needing to look at every example in the dataset are stochastic gradient descent and mini-batch stochastic gradient descent.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Stochastic gradient descent

- Stochastic gradient descent (SGD) uses only a single example (a batch size of one) per iteration. Given enough iterations, SGD works but is very noisy.
- "Noise" refers to variations during training that cause the loss to increase rather than decrease during an iteration.
- The term "stochastic" indicates that the one example comprising each batch is chosen at random.



Loss slightly fluctuates as the model updates its weights and bias using SGD, which can lead to noise in the loss graph.

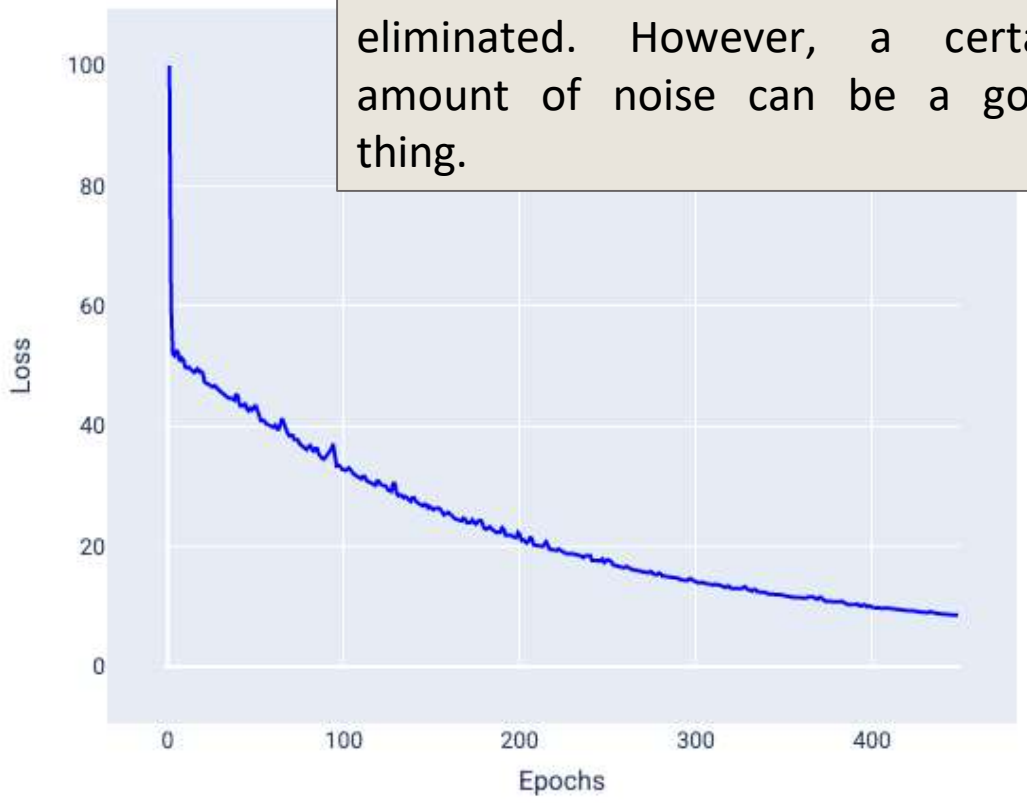
Stochastic gradient descent can produce noise throughout the entire loss curve, not just near convergence.

Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Mini-batch stochastic gradient descent

- Mini-batch stochastic gradient descent (mini-batch SGD) is a compromise between full-batch and SGD.
- For N number of data points, the batch size can be any number greater than 1 and less than N.
- The model chooses the examples included in each batch at random, averages their gradients, and then updates the weights and bias once per iteration.

When training a model, you might think that noise is an undesirable characteristic that should be eliminated. However, a certain amount of noise can be a good thing.



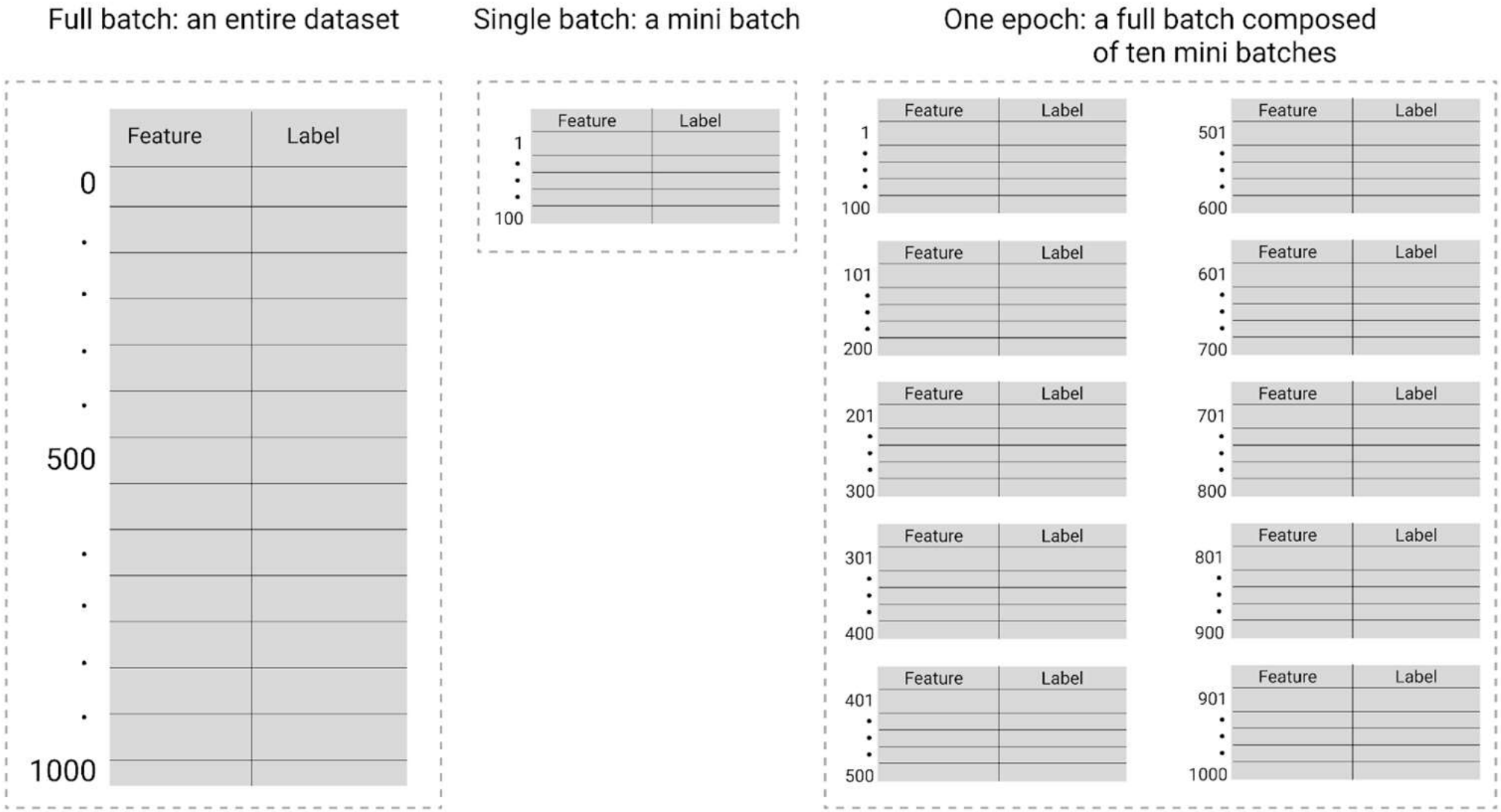
Source: https://developers.google.com/machine-learning/crash-course/linear-regression

Epochs

- During training, an epoch means that the model has processed every example in the training set once. For instance, a training set with 1000 examples and a mini-batch size of 100 examples, it will take the model 10 iterations to complete one epoch.
- Training typically requires many epochs and the system needs to process every example in the training set multiple times.
- The number of epochs is a hyperparameter that you need to set before the model begins training.
- In many cases, you'll need to experiment with how many epochs it takes for the model to converge.
- In general, more epochs produces a better model, but also takes more time to train.

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Epochs



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Epochs

This table describes how batch size and epochs relate to the number of times a model updates its parameters.

Batch type	When weights and bias updates occur
Full batch	After the model looks at all the examples in the dataset. For instance, if a dataset contains 1,000 examples and the model trains for 20 epochs, the model updates the weights and bias 20 times, once per epoch.
Stochastic gradient descent 1 example = 20 epochs 1000 x 20 = 20,000	After the model looks at a single example from the dataset. For instance, if a dataset contains 1,000 examples and trains for 20 epochs, the model updates the weights and bias 20,000 times.
Mini-batch stochastic gradient descent 1000 example / 100 = 10 10 x 20 = 200	After the model looks at the examples in each batch. For instance, if a dataset contains 1,000 examples, and the batch size is 100, and the model trains for 20 epochs, the model updates the weights and bias 200 times.

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THANK YOU

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